Exp 4a – Law of Constant Composition

INTRODUCTION

In this lab experiment, you will get to observe the physical properties of pure elements, and the physical properties of a compound formed from those same elements. By recording mass measurements along the way, and comparing to other students in the class, you will determine if copper and sulfur react in the same mass ratio each time.

BACKGROUND

Before modern atomic theory was developed, scientists relied upon careful mass measurements and visual observations for determining the composition of matter. In 1789, Antoine Lavoisier developed the "Law of Conservation of Mass", which states that the mass of an element before a chemical reaction is the same as the mass of that element after the reaction. This means that mass (matter) cannot be created nor destroyed in a chemical reaction. The "Law of Constant Composition" (also known as the Law of Definite Proportion) was developed in 1797 by Joseph Proust. The law of constant composition says that all samples of a particular compound will be made up of the same elements in the same proportion or ratio by mass.

It was not until later that the English meteorologist, and chemist, John Dalton formulated the first modern description of atomic theory in 1803, which helped explain why the Law of Constant Composition held true. Dalton proposed that each chemical element is composed of atoms of a single, unique type, and though they cannot be altered or destroyed by chemical means, they can combine to form more complex structures (chemical compounds). All samples of a particular compound are made up of the same number of atoms of each element within the compound. Since Dalton reached his conclusions by experimentation and examination of the results in an empirical fashion, this marked the first truly scientific theory of the atom.

Atomic theory has been revised over the years to incorporate the existence of atomic isotopes and the interconversion of mass and energy. In addition, the discovery of subatomic particles has shown that atoms can be divided into smaller parts. However, Dalton's importance in the development of modern atomic theory remains recognized.

In this experiment, a mixture of copper and sulfur will be heated to produce a copper and sulfur compound:

$$copper + sulfur \rightarrow copper\ and\ sulfur\ compound$$

More sulfur than necessary will be used to ensure all copper reacts with sulfur. Extra sulfur will react with oxygen from the air to produce sulfur dioxide gas, leaving only the copper-sulfur compound product behind. The mass of sulfur reacted will be determined by the law of conservation of mass (and knowing the mass of copper used and copper-sulfur compound made).

ADDITIONAL READING

- Law of Conservation of Mass
- Law of Constant Composition
- Dalton's Atomic Theory

SAFETY EQUIPMENT FOR LAB

- Goggles
- Lab coat

SUPPLY LIST

- Ring Stand
- Crucible
- Steel wool
- Bunsen Burner
- Gas hose
- Iron Ring
- Clay Triangle
- Crucible tongs
- Watch Glass
- Wire Mesh
- Flint Lighter (striker)
- Glass stirring Rod

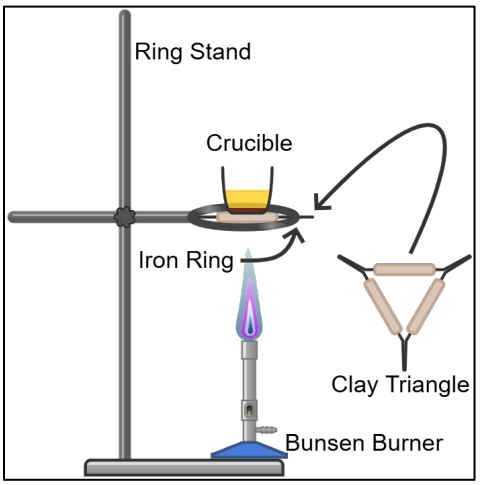
CLASSROOM SUPPLIES AND REACTANTS

These are shared supplies or reactants.

- Sulfur (solid)
- Copper wire
- Scoopula
- Fume Hood

PROCEDURE

Figure 1.
Bunsen Burner Setup



Cleaning the Crucible

Note: Do NOT collect any measurements prior to cleaning the crucible.

1. ☐ In the fume hood, setup the apparatus shown in Figure 1.
2. ☐ Obtain a crucible. If the crucible is dirty, use a small piece of steel wool to scrape any solids out of the interior, and wipe with a dry paper towel.
3. ☐ Place the crucible on the stand and heat over a blue flame until it glows slightly red-hot.
4. ☐ Use the crucible tongs to set the crucible on the wire mesh to cool to room temperature (about 5 minutes)
   ** Do not set a hot crucible directly on the countertop or on paper. It will burn the counter or paper.*

Collecting Data

Note: Do NOT skip the cleaning procedure, even if your crucible is new or appears clean.

5. ☐ Record the mass of the empty crucible. (Remember to *always* record all digits shown on the balance.)
6. ☐ Obtain a piece of copper wire. Note the "size" of copper wire obtained (small, medium, large). Coil the wire into a flat circle or spiral that lays flat at the bottom of the crucible.
7. ☐ Observe the color and tactile feel of the copper wire. Mark your observations on the report sheet.
8. ☐ Place the copper wire in the crucible. Record the mass of the crucible containing the copper wire.
9. ☐ On a piece of weigh paper, use a scoopula to obtain between 2-4 g of sulfur. Transfer the sulfur to the crucible so that the copper wire is completely covered.
10. ☐ Record the mass of the crucible containing the copper wire and sulfur.
11. ☐ Heat the crucible over a blue flame until all the sulfur is melted into a honey colored liquid. *It takes less than a minute!*
12. ☐ Using crucible tongs, move the crucible onto the wire mesh to cool for about 5 minutes.
    ** The melted sulfur will react with the copper during this time.*
13. ☐ Heat the crucible over a blue flame until all the excess sulfur has burned off. As sulfur burns, it will turn reddish black in color, eventually turning into sulfur dioxide gas. When it is done, the crucible will look dry and clean inside with only the coil of "copper-sulfur compound" remaining.

14. ☐ Use the crucible tongs to set the crucible on the wire mesh to cool. Allow the crucible to cool to room temperature (about 5 min).
15. ☐ Record the mass of the crucible containing the copper and sulfur compound.
16. ☐ Place the copper and sulfur compound on a watch-glass. It should slide out of the crucible easily.
17. ☐ Observe the color. While holding the compound over the watch glass, try to bend it back into a straight line. Make note of the tactile feel of the copper and sulfur compound.

CLEAN-UP AND WASTE DISPOSAL

1. ☐ Place the copper and sulfur compound in the solids waste container.
2. ☐ Wipe the inside of the crucible with a dry paper towel. Do NOT get the crucible wet.
3. ☐ Wash the watch glass with soapy water.
4. ☐ Put back all equipment (Bunsen burners, iron rings, etc.) back in the appropriate locations.

DATA AND REPORT – DUE ____ / ____ / ____ **SECTION _____**

* Show your work and keep track of significant figures. **DATE _____**

PARTNER _____

DATA

After Cleaning, Before Reaction:

1. Description of wire before reaction:
    a. Size of copper wire obtained: ____ small ____ medium ____ large
    b. The wire feels: ____ flexible ____ rigid ____ brittle
    c. The color of the wire is: ____ dark gray ____ light gray ____ metallic copper (shiny)
    ____ red brown (dull) ____ yellow

	Mass of…	**Mass (g)**
A	Crucible (empty)	
B	Crucible + copper	
C	Crucible + copper + sulfur	

After Reaction:

	Mass of…	**Mass (g)**
D	Crucible + copper and sulfur compound	

2. Description of wire after reaction:
    a. The wire feels: ____ flexible ____ rigid ____ brittle
    b. The color of the wire is: ____ dark gray ____ light gray ____ metallic copper (shiny)
    ____ red brown (dull) ____ yellow

CALCULATIONS

HOW	MASS OF…	MASS (g)
[B-A]	Copper wire (Initially)	
[C-B]	Sulfur (Initially)	
[D-A]	Copper and sulfur compound	
[B-A]	Copper reacted	
	Sulfur reacted	
	Excess sulfur	

1. Mass Ratio of "copper / sulfur" reacted based on your data (Divide): _____

MASS RATIOS OF OTHER GROUPS:

	Group A	Group B	Group C	Group D	Group E	Group F	Group G
Mass Ratio							
Cu Size (S, M, L)							

	Group H	Group I	Group J	Group K	Group L	Group M	Group N
Mass Ratio							
Cu Size (S, M, L)							

2. Average mass ratio of class (excluding your data): _____

3. Calculate the percentage difference of your ratio compared to the class average: _____

$$Percent\ Difference = \frac{Your\ Ratio - Class\ Ratio}{Class\ Ratio} \times 100$$

QUESTIONS

1. Compare the physical properties of the wire you observed before and after heating:

   a. How are they similar?

   b. How are they different?

2. Analyze the table containing mass ratios from other groups. Is there a clear trend between the size of copper wire used and the mass ratio obtained? According to the Law of Constant Composition, should there be a trend?

   Explain using complete sentences.